VIDYAPEETH

BATCH CODE: 12-AN-181MA-2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:

UNIT & MEASUREMENT

Lecture No. 17

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Today's Goal

Subtopic Vernier Calliper, Screw Gauge

Subtopic Zero error Calculation

Subtopic

$$\frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\frac{\Delta R_{eq}}{(3.75)^2} = \frac{0.\cancel{3}}{\cancel{360}\ 120} + \frac{0.\cancel{2}}{\cancel{1000}\ 500}$$

$$\Delta R_{eq} = \left(\frac{1}{120} + \frac{1}{500}\right) \times (3.75)^2$$

$$= [0.0083 + 0.002]\ 14.0625$$

$$\boxed{\Delta R_{eq} = 0.1448}$$

$$\% \text{ error} = \frac{\Delta R_{eq}}{R_{eq}} \times 100$$

$$\frac{0.1448}{3.75} \times 100 = 3.84\ \%$$

Vernier calliper

- $LC = \frac{n-m}{n}$

  $LC = 0.1\,mm$ or $\frac{1}{10}\,mm$.

- $10\,VSD = 9\,MSD$

- $l = MSR + n.(LC)$

  $MSR \rightarrow$ Main Scale Reading.

ZERO ERROR IN VC :-

$N \rightarrow$ No. of VSD (ordinary)

$n \rightarrow n^{th}$ div of VS coinciding with MS

1)

No-zero-Error ✓

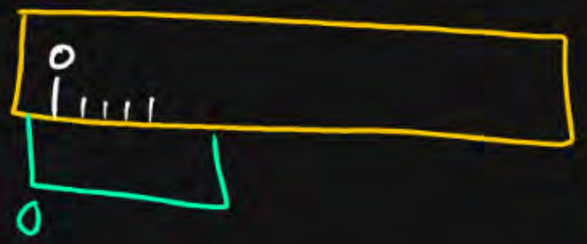
2) ☆

Positive zero Error

$PZE = \cancel{MSR}^{0} + nLC$

$\boxed{PZE = (n)\,LC}$

Correct Reading = Measured Reading − PZE

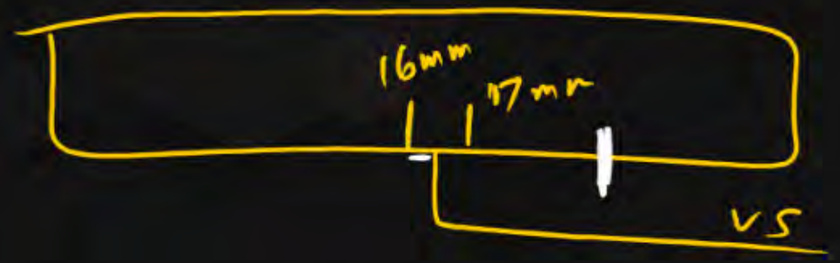
3) 

Negative Zero Error.

$NZE = (N-n)\cdot LC$

$\boxed{\text{Correct Reading} = MR + NZE}$

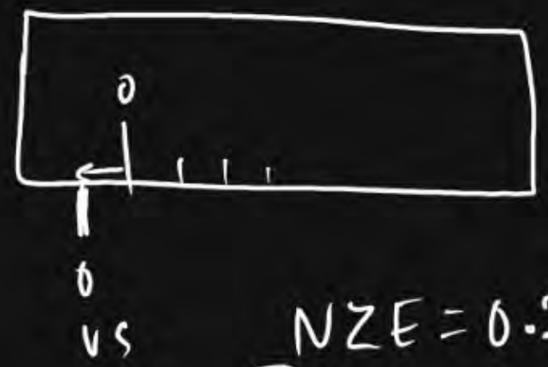
#



16
MSR + n(LC)
4 (0.1)

= 16 + 0.4
= 16.4 mm

16.4
16.2
16.6
16.



NZE = 0.2 mm

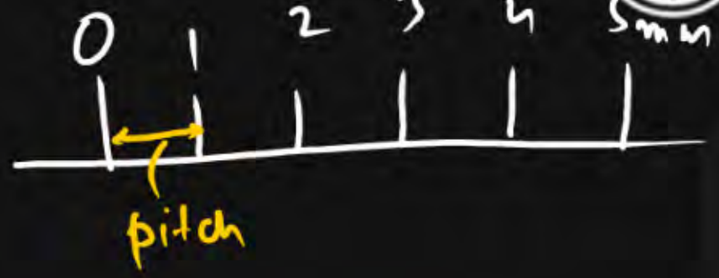
given :-

Correct = 16.4 + 0.2
= 16.6 mm

1.66 cm

SCREW GAUGE :-

$$LC = \frac{1}{100}\,mm = 0.01\,mm$$



① $$\boxed{LC = \frac{\text{Pitch}}{\text{no. of CSD}}}$$

BG
MG = 1m
NG = 0.72 m

Ordinary SG → no. of CSD = 100.

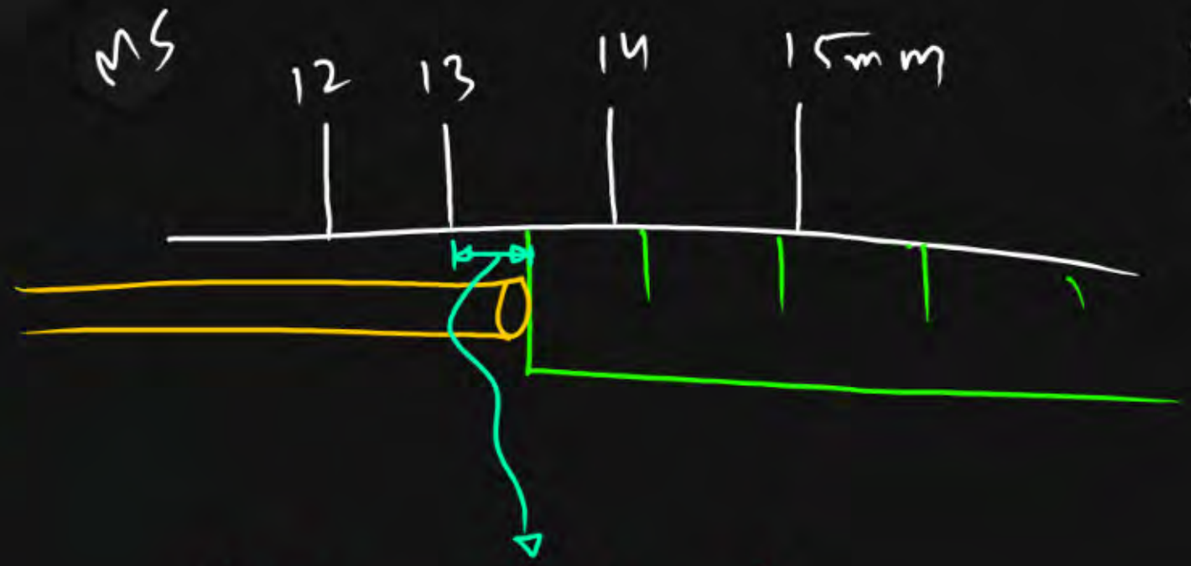
② $$\boxed{l = MSR + n.(LC)}$$

$16 + 4(0.1)$

$16.4\,mm + 0.2$

$\underline{16.6}$

$13 < l < 14$



MSR + n.(LC)

4 (0.1)

A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale.

The pitch of the screw gauge is :

[NEET-2020 (Phase-1)]

(1) 0.25 mm

(2) 0.5 mm

(3) 1.0 mm

(4) 0.01 mm

0.05 m

5 m

0.005 m

$$LC = \frac{\text{pitch}}{CSD}$$

$$\text{pitch} = LC \times CSD$$

$$= 0.01 \times 50 \text{ mm}$$

$$= 0.5 \text{ mm}$$

2016

A screw gauge gives the following readings when used to measure the diameter of a wire

Main scale reading : 0 mm

Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is **[NEET-2021]**

(1) 0.052 cm (2) 0.52 cm

(3) 0.026 cm (4) 0.26 cm

$l = MSR + n \cdot LC$

$= 0 + 52(0.01)$

$= 0.52\,mm$

$= 0.052\,cm$

$LC = \frac{pitch}{CSD} = \frac{MSD}{CSD} = \frac{1mm}{100} = \frac{1}{100}mm = 0.01$

✓ ✓ ✓ ✓

✓ — — —

✓ ✓ ✓ ✓

✓ ✓ — —

चेप्टर समाप्त

A 3

1 2,3 4 5,6 7 rule

1st R 2nd R.

Wed < P - 1h, C - 1h

SUN < B, P, 2, C

★★★★★

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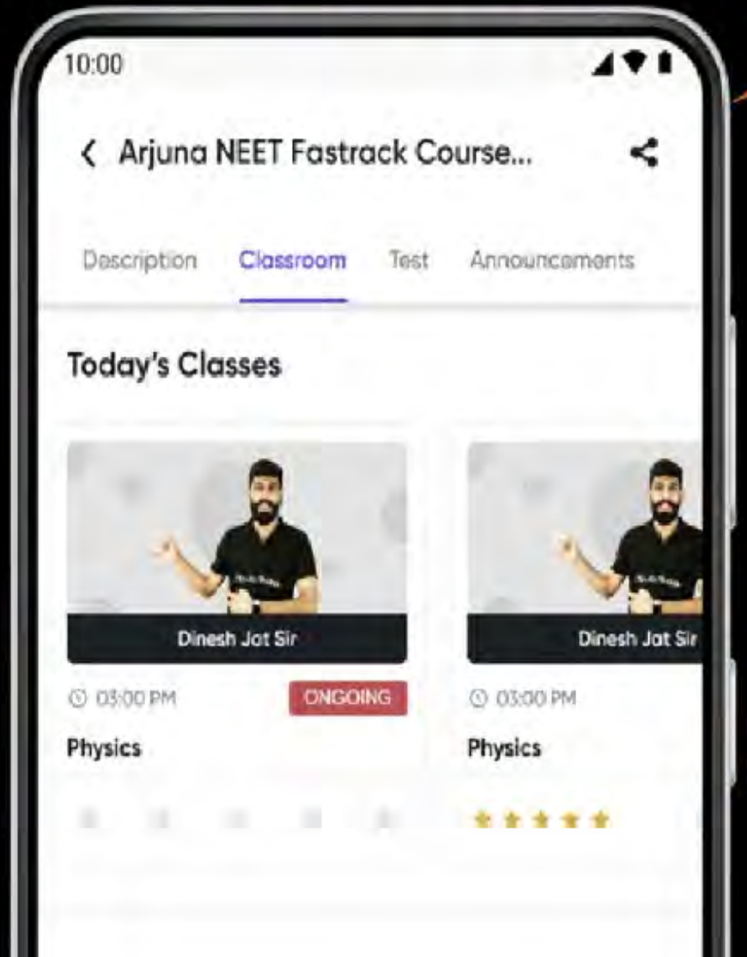
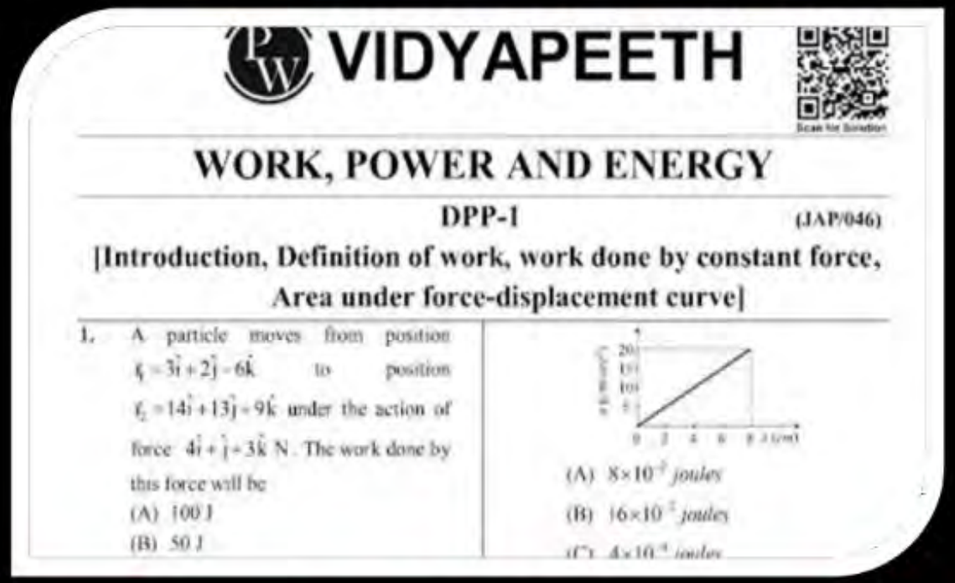


Solve the DPP and check Solution



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